

A Deep Learning Based Method for Plant Root Segmentation

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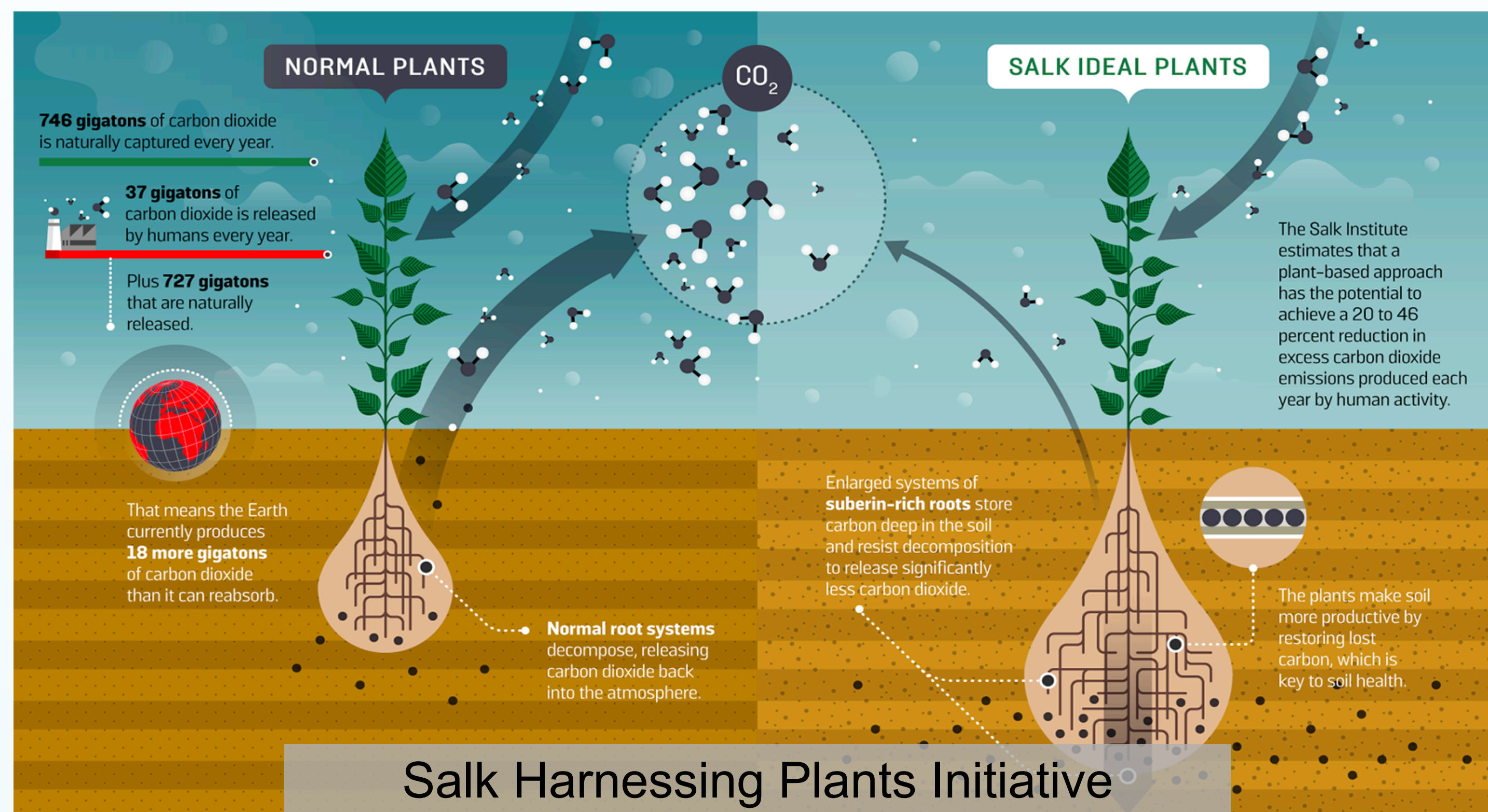
The Salk Institute, Waite Advanced Biophotonics Core, San Diego, California

Introduction

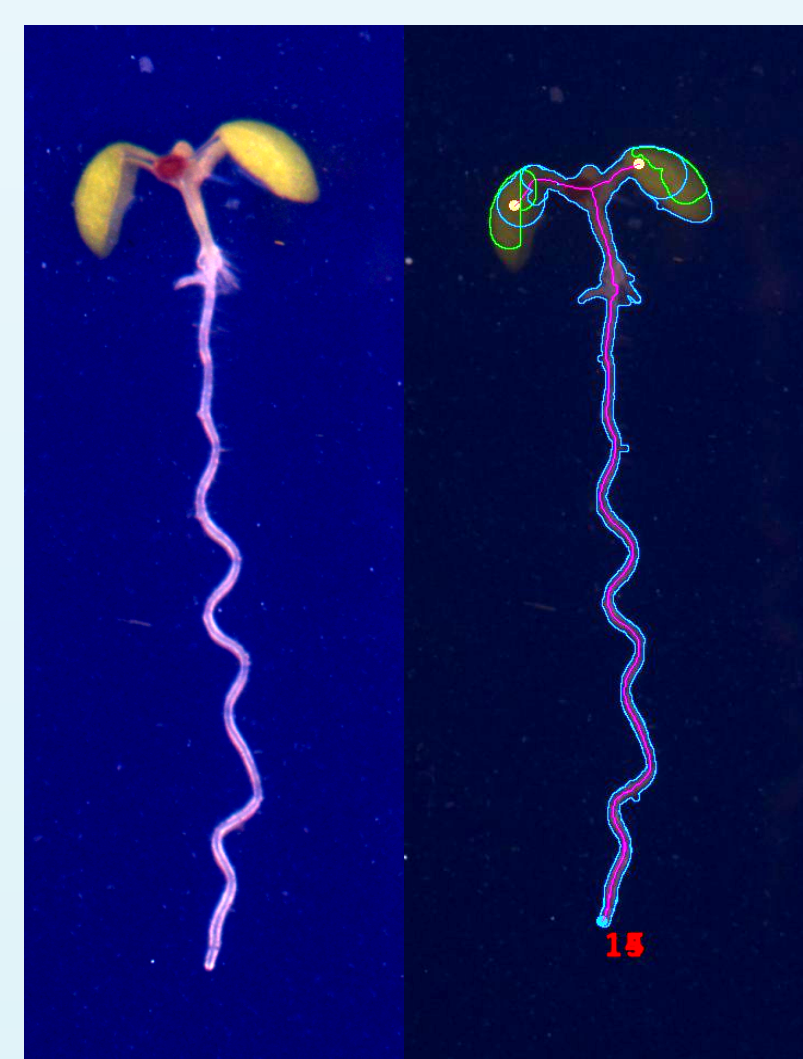
The Life Science Summer Institute is a program which provides the tools to high schoolers to conduct hands-on research with a one week boot camp and seven weeks of a paid internship at Salk. My work this summer was based on integrating deep learning into plant biology to trace plant root images.

Background

- A major challenge in biology is to relate the genotype with phenotype; particularly, in plant biology, the root is the subject of frequent observation because of its growth properties



- A large-scale phenotyping pipeline has recently been developed to acquire large data sets of plant root images: robotic techniques to plant and image plant roots in specific nutrient environments held in control chambers. There are 24 roots planted in a row-like manner in a lab plate and photographed by a separate camera over a 14 or 20 day period.
- Manually tracing roots can be very laborious, thus the Busch Lab, run by Dr. Wolfgang Busch, devised a Busch Lab Root Analysis Toolchain (BRAT) which is very effective for tracing simple root phenotypes.

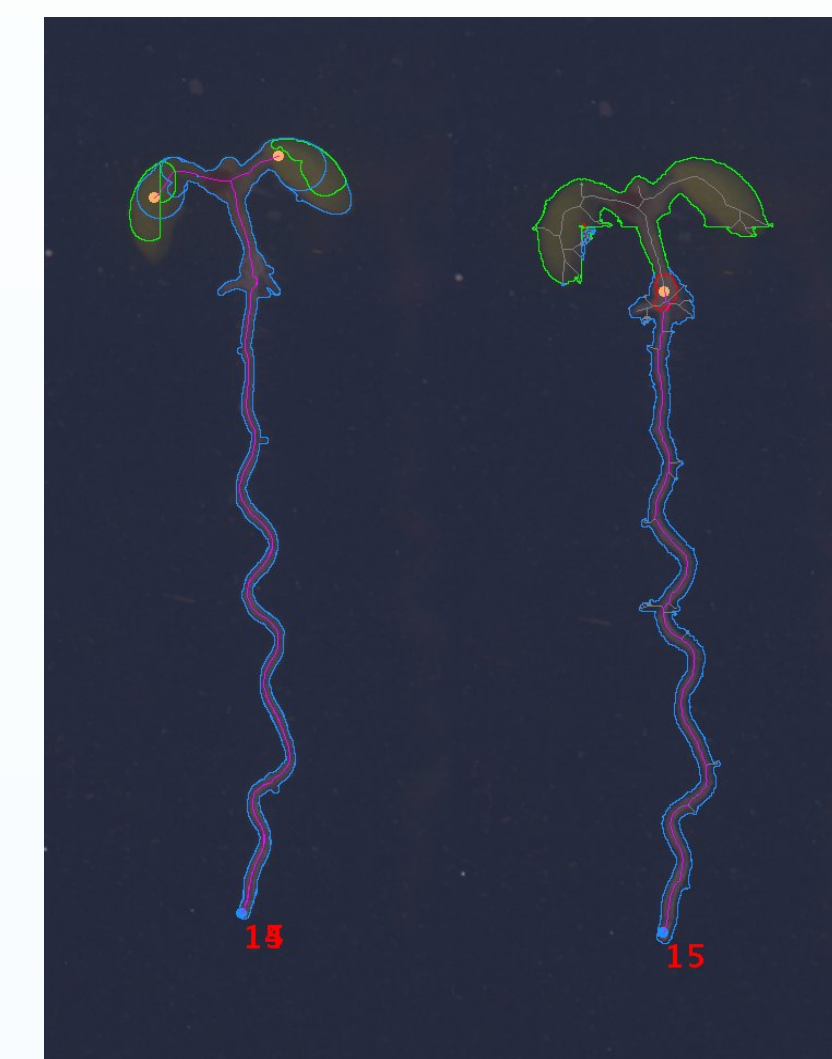


Original Image BRAT Version

- BRAT is ineffective in tracing complex root phenotypes and uses a relatively simple method to trace the roots. We turned to machine learning in order to trace complex root images and more accurate classification
- Machine learning uses algorithms to parse data, learn from that data, and make informed decisions based on what is learned

Deep Learning Solution

- Deep learning structures algorithms in layers to create an “artificial neural network” that can learn and make intelligent decisions on its own, mimicking the human brain
- We used deep learning to modify the BRAT toolchain, thus named deepBRAT



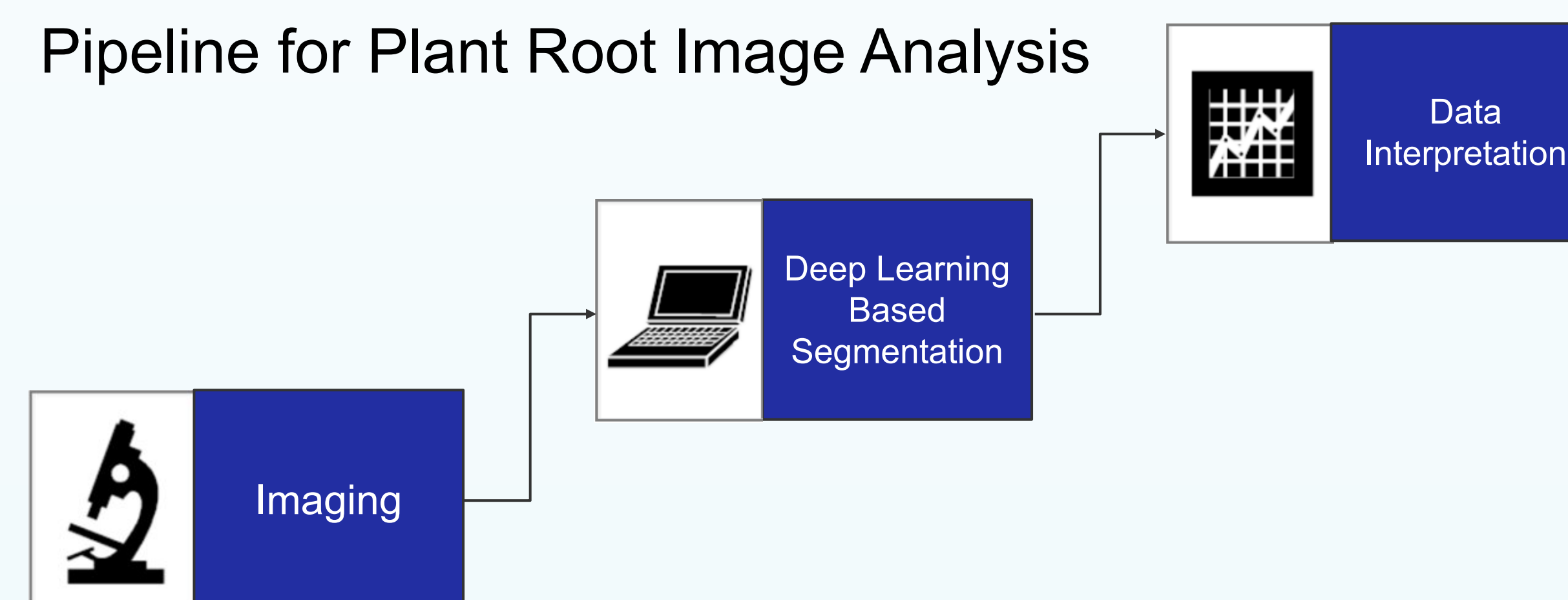
BRAT deepBRAT

Advantages of deepBRAT Method

- Significantly faster
- Detects primary and branched roots
- Improved classification
- Correctly identifies root position on lab plate

On the left is a comparison of the results of the BRAT segmentation versus the deepBRAT segmentation.

Pipeline for Plant Root Image Analysis

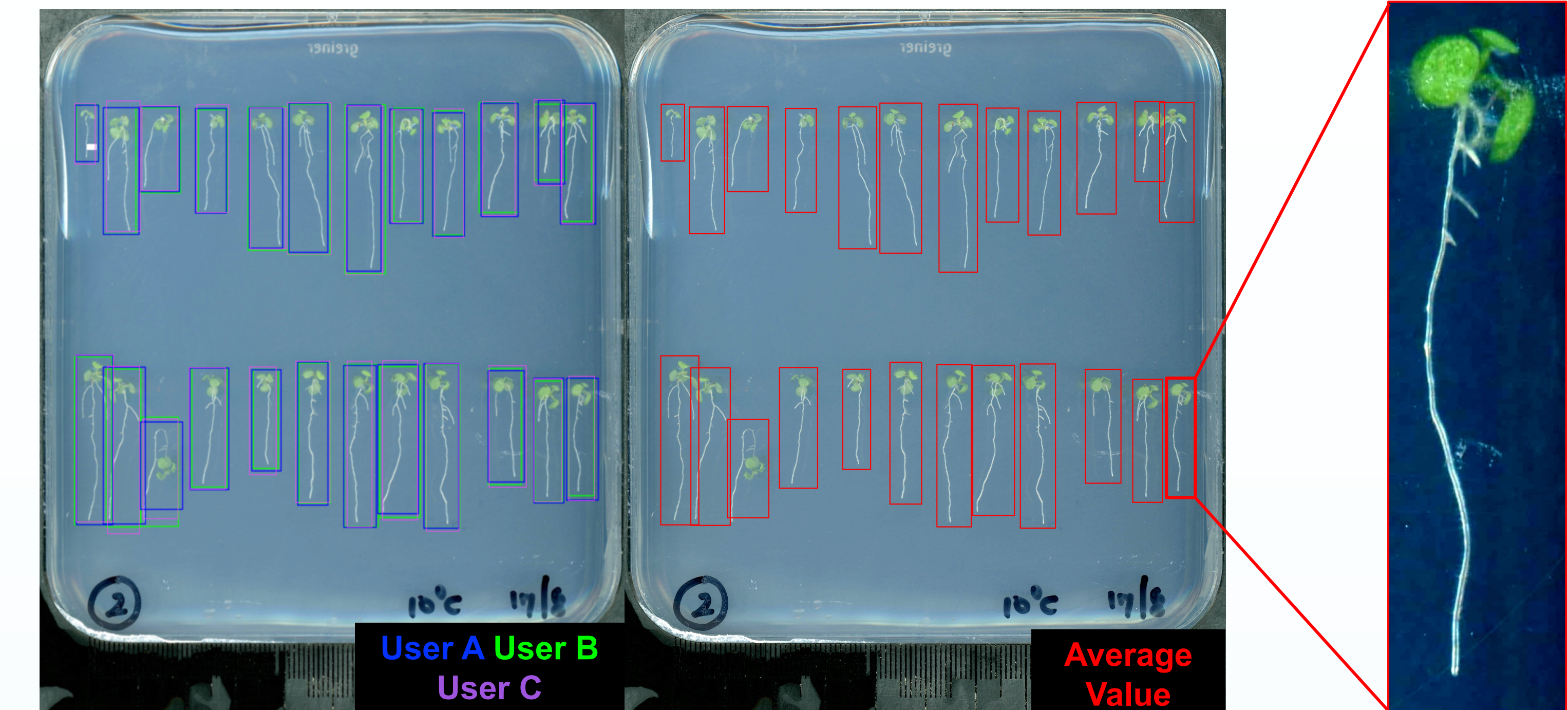


Collecting Ground Truth Material

- In order to optimize the deepBRAT model, we need to train it using Ground Truth, or the “perfect” data. Our method was to receive ground truth using crowdsourcing methods, in which citizen scientists would trace root images.

Using Zooniverse (Crowdsourcing website)

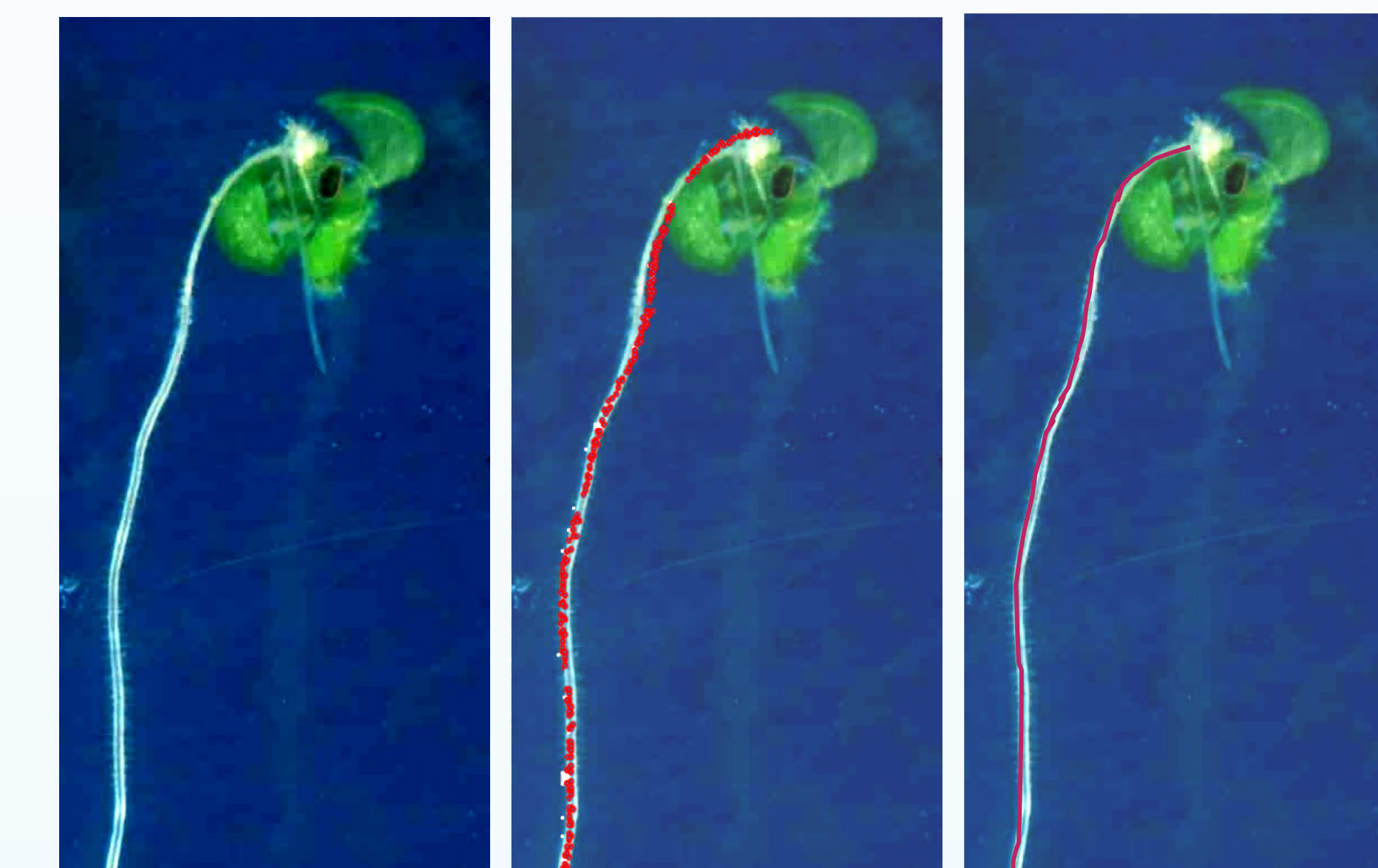
- Upload lab plate images to zooniverse using server
- Citizen scientists create bounding boxes on Zooniverse
- Average bounding boxes and individualize root data
- Upload individual root images back to Zooniverse server
- Citizen scientists trace primary and secondary roots



Lab plate overlaid with bounding boxes of three separate users

Lab plate overlaid with average value of three users' bounding boxes

Cropping root using ImageJ software



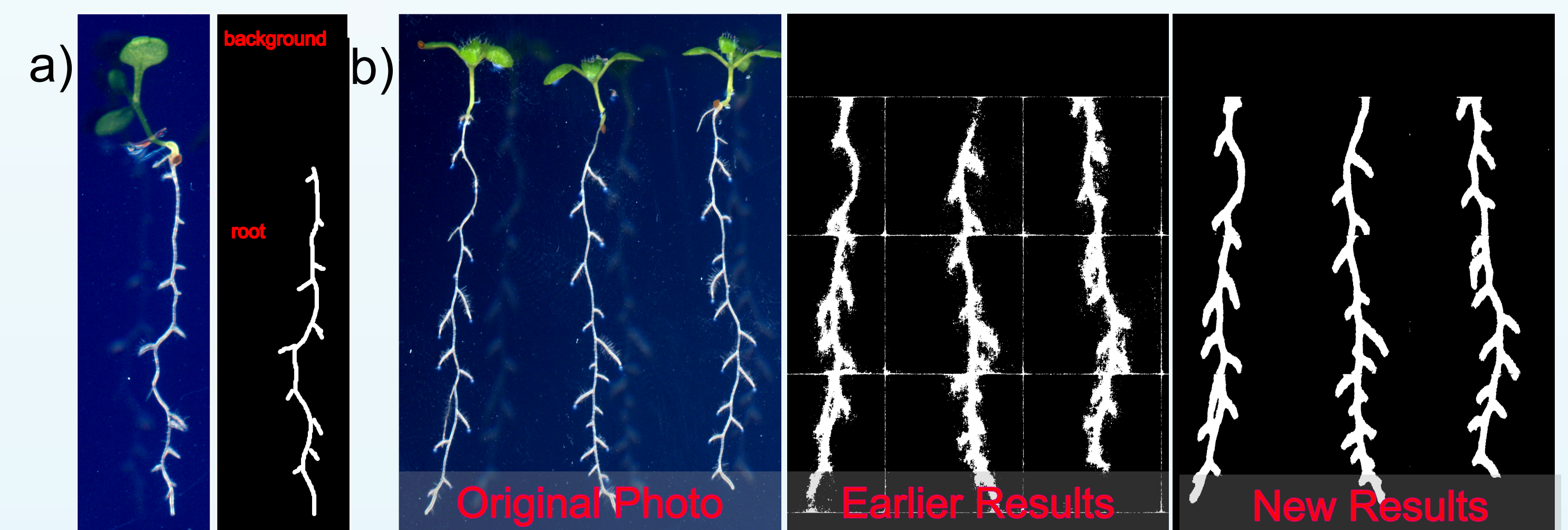
Original Photo

Point Cloud

Regression Line

After citizen scientists trace root image, points form a point cloud of over 300 points. In order to create tracing from these points, we use a B-Spline function which minimizes deviation between the original points and the final trace.

Conclusions



a) Training Pair for deepBRAT machine learning model

b) Predictions before and after more training pairs are used to optimize the model. Using my work this summer, the results of optimized model is shown to the right.

In the future, we hope to find a correlation between the morphology of plant roots and their environment nutrients

Acknowledgements

I would like to thank my mentor, Linjing Fang, the director of BPHO, Uri Manor, and my family for their persistent support of my work.